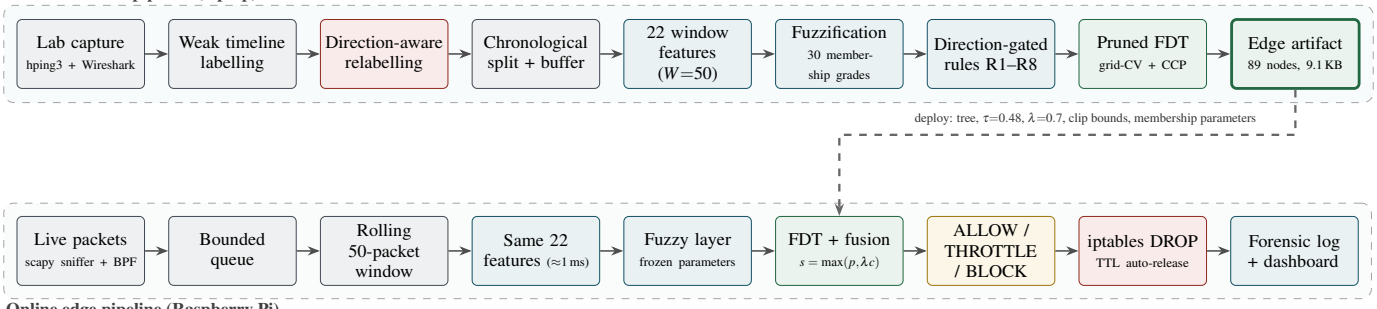

Offline research pipeline (laptop)
Lab capture
hping3 + Wireshark
Weak timeline labelling
Direction-aware relabelling
Chronological split + buffer
22 window features (W=50)
Fuzzification
30 member-ship grades
Direction-gated rules R1–R8
Pruned FDT
grid-CV + CCP
Edge artifact
89 nodes, 9.1 KB
deploy: tree, τ=0.48, λ=0.7, clip bounds, membership parameters
Live packets
scapy sniffer + BPF
Bounded queue
Rolling 50-packet window
Same 22 features (≈1 ms)
Fuzzy layer
frozen parameters
FDT + fusion
s = max(p, λc)
ALLOW / THROTTLE / BLOCK
iptables DROP
TTL auto-release
Forensic log + dashboard
Online edge pipeline (Raspberry Pi)